A Two-Layer Feasibility-Desirability Architecture for Personalized Intervention Selection: Design Rationale and Application to Human Spaceflight Countermeasures

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1. Introduction

1.1 The human spaceflight countermeasure selection problem

The spaceflight environment exposes astronauts to microgravity-induced multisystem deconditioning, leading to physiological effects including bone density loss, muscle atrophy, cardiovascular deconditioning, and body fluid redistribution [1]. Additionally, it is an isolated, confined, extreme (ICE) environment that takes a significant toll on circadian rhythms, psychosocial well-being, and interpersonal relationships. *Human Spaceflight Countermeasures* (HSC) are the tools used to mitigate the physiological effects, primarily exercise protocols but also nutritional and pharmacological strategies.

Mission planners and spaceflight medicine experts are entrusted with the task of determining which specific countermeasure protocols should be prescribed to each individual crewmember, taking into account a set of constraints including the individual crewmember's unique physiology, their current state, finite mission time, highly constrained resources, and multiple systems to protect [2]. This is a *selection-under-constraint* optimization problem that is well-suited to be supported by predictive data analytics methods. The factors that make personalized countermeasure prescription difficult to optimize can be largely summarized as four discrete dynamics: outcomes are uncertain, multiple systems compete for limited protocol time and resources and cannot all be optimized simultaneously, different individuals have unique needs, and outcomes are mission-critical and therefore planners are accountable for the consequences.

Human spaceflight countermeasure selection is an instance of a broader class of optimization problems that seek to select interventions to steer an individualized system toward a desired state under uncertainty and competing constraints. The architecture presented in this report aims to provide an elegant solution to this class of problem, providing mission-critical decision support not only for HSC but for a wider variety of use cases.

1.2 The shortcomings of current approaches

Current HSC practice is demonstrably insufficient in preventing multisystem deconditioning in astronauts, as evidenced by a 2023 study evaluating the effects of exercise countermeasures on multisystem function in a cohort of 46 astronauts on long-duration spaceflight missions [3]. The researchers found substantial inter-individual heterogeneity in changes to pre- to post-flight state, ranging from -30% to +5%. It was concluded that an estimated 17% of astronauts would experience performance-limiting deconditioning if current HSC practice were to be implemented in future missions [3]. Additionally, it was found that maintaining or increasing the dose of current HSC protocols still coincided with continued post-flight deconditioning, evidence that simply escalating dose within current practice does not straightforwardly resolve the problem. The researchers note some factors making this a difficult problem to solve, including hardware inconsistencies, difficulty in comparing different results from individual astronauts, and uncertainties regarding transferability of exercise on earth with equivalent protocols under

microgravity [4]. The same study also detailed the HSC selection process for European Space Agency (ESA) astronauts, in which human experts analyze individual astronauts' pre-flight training data (standard ESA fitness test [5]) and regular re-prescription of exercise countermeasures based on a monthly submaximal cardiovascular capacity test (spiroergometry) along with self-reported reports on exercise intensity and personal feedback.

Another important aspect of HSC selection is recognition that intrinsically, it is a trade-off analysis among physiological and resource expenditure objectives which, under current practice, is resolved implicitly. It has been documented that the combined resistive and aerobic exercise protocol prescribed on the International Space Station raises total energy expenditure by about 15% and drives an overall negative energy and nitrogen balance that, counterintuitively, is detrimental to muscle mass. Prescribing resistive-only protocols helps preserve fat-free mass and lower energy expenditure, but is inadequate against cardiovascular deconditioning [6]. This tension is compounded by resource constraints – because exercise duration and intensity are proportional to energy requirements, food transport, resupply/production, and storage necessitates a delicate equilibrium vitally important to mission planners. Additionally, specific HSC protocols may have conflicting effects in the astronaut, where an intervention that supports one system may have a negative effect on another, such as through exercise-induced elevations in intrathoracic and intracranial pressure leading to neuro-ocular risk, intense exercise increasing injury risk to spaceflight-weakened joints, and optimizing for minimum effective exercise dose may counteract the psychological well-being effects of being active for longer periods [7], [8]. Across multiple axes, the field lacks an explicit definition of what constitutes an acceptable multidimensional outcome. The weighing of objectives against one another is performed via expert judgement, and is done without predictive data analytics methods providing mission planners with explicit representations to interrogate.

NASA's exercise protocols are clearly specified, bounding exercise intensities to fixed bands as percentages of individuals' pre-flight performance metrics, meaning that individualization consists of scaling standardized intensity bands to a measured baseline [9], [10]. There is currently no computational model in place for predictively characterizing what physiological states are actually attainable/likely for individual astronauts under current or hypothetical protocols, no explicit multi-objective optimization for evaluating multisystem trade-offs, and no well-defined standards for comparing potential interventions before implementation.

1.3 Contribution and scope

The contribution of this work is the algorithmic architecture for a decision-support tool that aims to address problems similar in class to the HSC context presented above. It is structured as a two-layer model that represents, separately, what an intervention can potentially achieve and how desirable each achievable outcome is, combining the two layers when queried to return an empirical assessment of competing objectives. The first layer, *feasibility*, characterizes the range of states a system can reach under a candidate intervention, along with the uncertainty associated

with each potential state. The second layer, *desirability*, scores every possible state according to how well it serves each objective under specified conditions, while preserving the respective contributions of each variable. Coupling the two yields, for each candidate intervention, not a single predicted outcome but a distribution of outcomes weighted by how likely and desirable they are, returning a set of non-dominated interventions to the human decision-maker to review before selection. The contribution thus lies in independent computation of these two layers such that when coupled they may provide unique insight impossible to generate using a monolithic computational method.

Stated in domain-neutral terms, the HSC-prescription context presented earlier is simply an instance of a general class of problem. The computational crux is to select interventions whose potential outcomes are simultaneously *reachable* (attainable via the intervention within a defined time horizon) and *desirable* (valued across competing objectives). Although HSC prescription exhibits this structure, it is not the unique example case. The same pattern recurs across domains; therefore, the architecture shall be developed as a domain-agnostic solution to this class of problem, treating the system's state variables, interventions, and objectives as configurable inputs.

The scope of this report is deliberately bounded to the decision architecture itself. In scope are the two layers, the mechanism of their coupling, the query classes the model answers, and the controls by which an operator may direct the analysis. By contrast, the downstream and upstream components on which the algorithm depends but to which does not directly contribute are treated as black-boxes. This includes data acquisition, storage, transmission, and feature engineering. The contribution is established at the architectural level with in-depth analysis of each important design decision, and a blueprint for the development of a working proof-of-concept software model. The aim at this stage is to show that the architecture is computationally realizable and that it can produce the promised decision-support output.

2. Architecture Concept

2.1 Governing ideas

The architecture rests on a single important observation: the questions "what can this intervention accomplish?" and "how good would that be?" are about different things, and trying to answer them simultaneously makes the problem intractable. The former is a question about the individual system and its dynamics, while the latter is about values, objectives, and evidence. Nothing requires them to be answered via the same computation, and seeking to do so actually loses a lot of valuable information.

Consider the real role of an HSC intervention. Prescribing a resistive exercise protocol does not make an astronaut's bones stronger, rather it reshapes the region of marker space that the astronaut is likely to occupy by a certain time horizon, shifting where the probability mass sits and how widely it spreads. Since the individual response varies substantially, the real description of

that effect is a distribution over reachable states rather than an exact predictable point. Desirability is how good a particular state is for an individual, and feasibility is where that individual's potential lies. To make an informed prediction, the two are brought together only when a decision must be made, and decisions are fickle things dependent on changing preferences, information, and contexts. To be flexible, it is important to not assume you know what the decision-maker will want, but rather give them as much information as you can an let them handle the rest. And, for critical missions, it really is best to leave decisions up to the experts, and not let computational models or anything else bias their decision. That is the rationale for returning a set of good choices as output, instead of an "optimum intervention" that gets spit out without considering all aspects of the situation.

The algorithmic architecture presented in this work is not a standalone tool but a component within a larger digital twin. A decomposition of a medical digital twin into 5 stages (acquisition, management, analysis, prediction, and interpretation) is adopted here as a conceptual scaffolding [11], [12]. *Acquisition* and *management* are out of the scope of this report, the architecture here assumes data has been collected, cleaned, standardized, and integrated. *Analysis* is where personalization occurs. Both layers are tuned to the individual from baseline measurements and updated as new data arrives. *Prediction* is the feasibility model. For each candidate intervention it propagates the personalized state forward, returning a weighted particle cloud over reachable states. *Interpretation* is comprised of the coupling and ranking stages, where the feasibility distribution is expressed through the desirability field; and ranking applies the decision-maker's preferences as inputs and returns a non-dominated set. This is where prediction becomes a recommendation.

The two layers are intentionally specified independently, and the architecture logic is developed to keep them that way. Independence here means three things. First, *separate provenance*: the feasibility and desirability models are built from different bodies of evidence, gathered by different means, and neither is derived from the other. Second, s*eparate revision*: either layer may be corrected, recalibrated, or replaced without affecting the other. Third, *coupling at query time*: the two are only combined when a decision is actually being made, in order to provide the decision-maker with as much unbiased information as possible so the best intervention may be chosen.

2.2 Architecture description

Figure 1 illustrates the two-layer architectural concept over a representative two-dimensional state space. The axes are two state variables; one may think of them as biomarkers in an HSC context. The *desirability field* is the continuous green-to-red gradient, where every point in the space carries a value, and the gradient is graded rather than bounded. There is no hard line separating acceptable from unacceptable, only a landscape that worsens in some directions faster than in others. The gradient's high-desirability center is often the individual's reference configuration (especially in the spaceflight context where the vast majority of physiological

alterations are deleterious), but it could in theory be anywhere. The asymmetry reflects the per-variable penalty structure. The *feasibility distribution* is represented by the set of nested contours. Each contour encloses the region the system is likely to occupy at a given horizon under one candidate intervention, with the innermost representing the nearest horizon and the outer ones progressively later. Although they appear bounded in the figure, they are usually probability clouds with soft gradient edges, reflecting the continuous nature of physiology. The distribution begins concentrated at the current state and spreads as the horizon extends, reflecting cumulative uncertainty. It also drifts, because the intervention reshapes where the probability mass settles. Superimposing the two is the coupling. The figure makes visible what the pushforward computes: how much of the distribution mass sits in unfavorable territory, how much has drifted into the "red", and in what direction.

What the figure cannot show should be stated explicitly. First, it depicts a single candidate intervention. The architecture produces one such landscape per candidate, and the decision is a comparison across them: the figure shows one instance of the object being compared, not the comparison itself. Second, the real state space has as many dimensions as there are tracked variables, and desirability is a vector rather than a color. The two-dimensional rendering is a projection for readability.

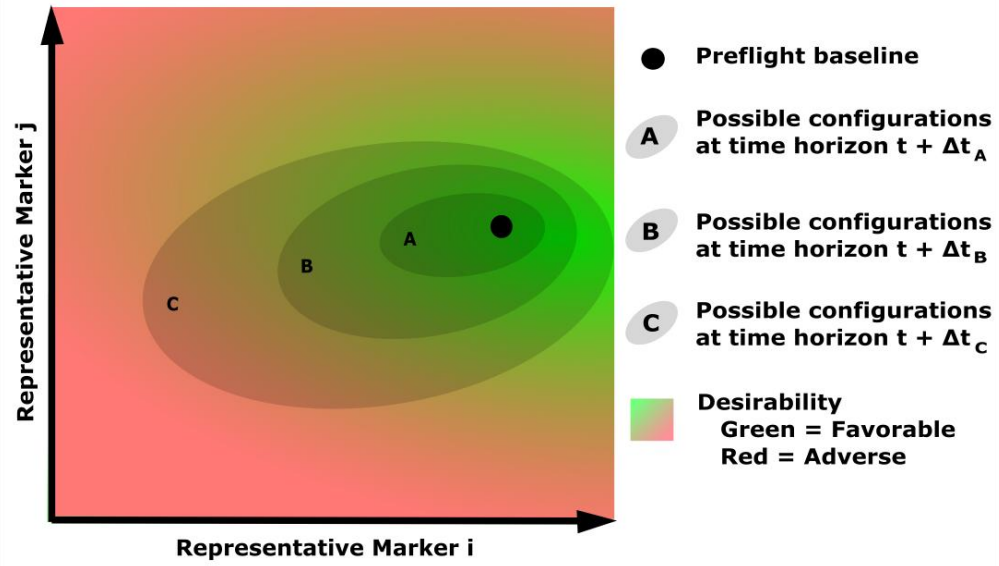


Figure 1. Conceptual illustration of the feasibility and desirability models over a two-marker state space. Axes and values are schematic, not computed.

3. State of the art and positioning

3.1 Novelty and component sources

The architecture developed in this work is mostly assembled from established components drawn from several distinct bodies of work, and it is worth stating plainly at the beginning what these components are and where they come from.

- The representation of system state attainability as a reachable state or distribution is drawn from **control theory**.
- The comparison of candidate actions by their full outcome distributions is drawn from **stochastic multi-objective optimization**.
- The personalization of a model from a population-level prior to an individual-level posterior is standard **Bayesian inference**.
- The final output being a set of evaluated intervention options for a human operator to choose between is common in the developing field of **digital twin modeling**.

The contribution's central novelty lies not in the origination of these individual concepts, but instead in the architectural combination of distinct techniques. The components listed above originate in fields that infrequently intersect, and this work seeks to draw them together into a single coherent framework for intervention selection. Specifically, there are 3 novel architectural characteristics that accomplish this goal:

1. Feasibility and desirability are specified independently from separate sources.
2. Feasibility and desirability are coupled only when a decision is queried.
3. Decisions are resolved by evaluating a supplied set of candidate interventions rather than searching for an optimum or learning a policy.

In addition to the algorithmic architecture, there are two supplementary novel contributions that a thorough search of the literature finds to be without mature precedent in the forms presented here. First, the works contributes a characterization of the class of problems the architecture addresses: intervention selection for an individualized, uncertain system under competing objectives where what is reachable and what is desirable arise from separate sources. Naming this class is what grounds the architecture's domain agnosticism, allowing the HSC use-case to be understood as its first instantiation rather than exclusively defining its scope. Second, the components drawn from existing domains are adapted such that they optimally serve the defined problem class. The desirability model, the feasibility model, and the coupling mechanism are each developed as custom solutions rather than used off the shelf.

3.2 Positioning against existing foundational approaches

This section justifies each design decision of the architecture. For each component, the requirements are stated, established alternatives from the literature are evaluated against them, and the approach adopted in the architecture is specified. Each method heading is marked with its outcome.

3.2.1 Value assignment over the state space

The desirability model is a purely *evaluative* operator, assigning value to every point in the state space without generating or forecasting any state itself. Its value may be variable over time, but time-dependence is strictly evaluative and not a propagated dynamic. It must represent each objective separately, keep component contributions un-aggregated, and contain evidence-based risk profiling for each variable. Several mature practices exist in the literature for assigning value over a multidimensional space, the model's position is defined by its relationship to each:

Multi-attribute utility theory (rejected). Multi-attribute utility theory is a decision support technique for when a decision-maker must choose from among a limited set of options, each with its own specific attributes, associated uncertainty, and preferences. Relative weights are assigned to the various attributes and aggregated into a formal model representing the overall evaluation of each option, after which the evaluations may be compared and ranked [13]. Although this is a simple and effective framework for determining relative desirability of various states, it is not robust enough of a method to faithfully mathematically represent the complex multidimensional desirability landscape necessitated by the algorithm presented in this work. Additionally, the early aggregation of variables restricts inspectability of choice logic in such a manner that defeats much of the purpose of this work's contribution.

Markov decision process (rejected). As the foundation of most reinforcement learning algorithms, Markov decision processes are formal models used to define a decision environment as a series of choices each with its own subset of outcomes, with associated probabilities and cost. They are structured as stochastic sequences of events in which the probability of each event depends solely on the system's current state, rendering them memoryless. Solving a Markov decision process produces a value function that assigns to each state the cumulative reward obtainable from it under an optimal policy, and this value function can be understood as a sort of desirability landscape over state space whose value at each point reflects the relative reward [14]. Although a powerful tool in machine learning, this method is ill-suited to act as the mechanism for the desirability model for two reasons. First, the reward function returns a single scalar, aggregating into one value instead of preserving the individual components of each state's desirability. Second, the function is computed using relative probabilities of arriving at any state, entangling reachability into the desirability model and thus directly undermining the requirement the two layers must be independently specified.

Derringer-Suich desirability (partially adopted). Originally designed to optimize industrial product development, this so-called *desirability function* converts multiple, conflicting response variables into a unified, dimensionless scale ranging from 0 (unacceptable) to 1 (ideal). Each response is passed through a graded function that returns a desirability of 1 when the response is exactly at its target, 0 if it falls outside of acceptable limits, and intermediate values in between. The specific shape of the function is chosen to reflect the shape of the response, whether it should be maximized, minimized, held at a target, and how steeply the value drops on either side [15]. This per-response construction is precisely what the proposed desirability model requires. Each variable in the desirability landscape can be independently bound, represented asymmetrically, and anchored to a unique reference. This flexibility captures the structure of physiological and other complex data types more faithfully than a monotonic penalty would. This aspect is adopted into the model. Despite this being well-suited, the fact that Derringer-Suich desirability function collapses the individual objectives into a single scalar means that it cannot be adopted entirely, but will rather serve as a partial inspiration for the final desirability model design.

Non-scalarizing methods (adopted). All of the previous examples shared a common aspect that was rejected from inclusion in the desirability model, namely early aggregation of component variables to a single scalar. There is another established family of approaches that leaves component variables in vector form, designed to preserve multi-objective structure and stochastically represent uncertainty throughout the analysis [16]. The desirability model adopts this strategy: the per-objective desirabilities are represented as a vector, one value per objective, and are never combined into an index value. This functions to keep each variable's value distinct and the trade-offs between objectives intact, leaving the comparison of states and interventions to the downstream coupling and ranking phases in order to maintain traceability and explainability to the greatest possible extent.

3.2.2 Forward reachability under uncertainty

The feasibility model is exclusively a *forward* operator and returns a purely *descriptive* output. It computes no optimum value and presupposes no target configuration. Its output is a probability measure over all reachable states, deliberately distinguishing its scope from the target-conditioned formulations that dominate reachability analysis and control theory literature, situating it as a method for *forward stochastic state characterization*. There are several principal adjacent method families, and the model's position is defined by how it differs from each:

Hamilton-Jacobi reachability (rejected). The Hamilton-Jacobi method is often used in control theory to determine the set of states that can possibly reach a target within a defined time horizon. It works by delineating a target zone (to be reached or avoided) and a cost function relative to the target, and assigning a value to every point in state space based on its distance from the target zone. It then, for a specified time horizon, can return the set of points from which it is possible to reach the target [17]. Although this is useful in many cases, especially in safety critical contexts where operators want to compute the dynamics of worst-case scenario situations, it is not

a good fit for the proposed digital twin architecture because it does not assign relative likelihoods of states within the reachable set, and also goes beyond the scope of the feasibility model by setting a target instead of being purely descriptive.

Kalman filter variants (rejected). The linear Kalman Filter and its nonlinear variants track the changing state of noisy dynamic systems over time by continuously blending predicted values with actual measurements, while assuming the noise has a Gaussian (bell curve) shape. The basic Kalman filter is limited to linear systems, but there are variants such as the Extended Kalman Filter and the Unscented Kalman Filter which are commonly used to model nonlinear dynamic systems [18]. The architecture in this work is designed to handle multimodal distributions that cannot be assumed to be Gaussian, rendering Kalman Filter variants suboptimal for this use case. A method is needed that can handle non-Gaussian noise.

Set-propagation methods (rejected). This family of reachability techniques work by representing the possible future state of a dynamic system by bounding a zone with a geometric shape, and propagating that shape over successive time steps. The shape can take many different forms based on data structure and computational budget (symmetrical zonotopes, support functions, ellipsoids, Taylor models, etc.), and can easily scale to high-dimensional data. Each instantiation of one of these propagation shapes represents a snapshot set of potential states at the given time horizon, which can be stacked along the time axis to create a "tube" shape called a *flow-pipe* representing the set of all possible states over a specified time interval. This method is often used in safety critical systems because if the flow-pipe never intersects with any pre-defined unsafe states, operators can claim with certainty that those states are not possible to reach [19]. Although forward-propagating and descriptive, this is not a perfect fit for this work's architecture because it returns a set of reachable states represented with equal likelihoods, not a probabilistic distribution.

Monte Carlo particle methods (adopted). Particle methods represent the state of a stochastic dynamic system via a large collection of weighted samples referred to as *particles*, each denoting an individual hypothesis about the system's state. Particle density in a region of state space approximates the probability of the system occupying that configuration. The state is propagated forward by drawing each particle through the system dynamics, so that the evolving cloud of particles traces how the probability mass spreads, shifts, and concentrates as the time horizon extends. It should be distinguished from the closely related *particle filter*, which augments forward propagation with a correction step that reweights particles against incoming observations. This is unneeded since the feasibility model projects forward from a personalized state rather than tracking one against new data (no observations during forward propagation). Because particles are simply forward-simulated rather than fitted to any assumed function form, the method places no restriction on the shape of the resulting distribution: it represents multimodal and non-Gaussian distributions just as naturally as unimodal ones [20]. It presupposes no target, computes no optimum, makes no Gaussian noise assumption, and its output is a genuine probability measure over reachable states rather than a bounded set of configurations carrying no probability measure. Its principal limitation is computational, a faithful approximation of the reachable distribution may

demand a large particle count, a cost which grows with the dimensionality of the state space. This is of particular relevance since the full particle population is intentionally carried through the model to the coupling stage, rather than having early dimensionality reduction applied. The feasibility model nevertheless adopts a Monte Carlo forward-propagation representation, considering it to be the closest representation of dynamic, multimodal, multidimensional, stochastic data.

3.2.3 Combining reachability and value

Coupling is the operation by which the outputs of the independent feasibility and desirability models are combined. It takes the reachability distribution produced for a candidate intervention over the specified horizon and the desirability field defined over the state space, and merges them into a statement of how "good" that intervention's attainable outcomes are. In order to achieve the stated goals of the system, three conditions are derived that distinguish the architectural requirements from conventional control theory. First, the coupling mechanism must **preserve structure**: it must not collapse the disparate objectives into a single score, nor reduce the distribution of possible outcomes to a single expected value. Second, it must **combine via evaluation**: this is intentionally *not* an optimization function, the coupling mechanism shall compute no optimum value and search no set of potential interventions, but rather simply evaluate the value of what a given intervention can reach. Third, it must **maintain objectivity**: no risk attitude, mission preference, or any other subjective judgement enters here. The risk tolerance and objective weightings that ultimately shape a decision shall be applied downstream; therefore, the direct output of the coupling mechanism must be an unbiased characterization of each intervention's outcomes. These requirements are restrictive and, as shown in the following literature analysis, the most established statistical coupling approaches from control theory and related fields do not satisfy them. The approach that is ultimately adopted is deliberately the simplest.

Expected-value coupling (rejected). The most immediate (and seemingly most intuitive) way to combine the two layers is by integrating the desirability field against the reachability distribution to obtain, per intervention, the expected desirability of its outcome. This is well-defined and simple, and it correctly weights good outcomes by how likely they are to be reached. Its main defect in the context of this work's system is that it reduces the entire outcome distribution to its mean. A protocol that usually has a highly desirable outcome but occasionally fails catastrophically, and a protocol that has reliably mediocre results may share the exact same expected desirability despite not being equivalent decisions [21]. This method's averaging mechanism discards precisely the spread of outcomes on which risk assessment depends, rendering it practically unusable for mission-critical decision making. Applied to a vector-valued desirability it is even worse, as a single expected value per objective (or an aggregated value across objectives) erases the distributional decompositional structure that allows operators to make informed decisions regarding risk. Expected-value coupling is therefore rejected from the architecture.

Chance-constrained coupling (rejected). Chance-constrained optimization is a method for determining optimal actions in uncertain systems. It works by defining a fixed probabilistic rule and an objective value to be optimized, and solves such that the defined uncertain constraints are satisfied with at least the specified probability. It casts feasibility as a probability threshold that the optimal solution must meet [22]. If implemented in the feasibility-desirability architecture presented here, it would define a hard desirability target region and seek to find the optimal intervention protocol(s) to get there within a specified minimum probability constraint. This does not satisfy the requirements for several reasons: it needs desirability to be expressed as a hard, bounded target region rather than a graded field, it is an optimization returning a single "best" intervention rather than a characterization of all candidates, and it fuses feasibility and desirability within a single constrained optimization variable rather than preserving their separability and internal components. For these reasons chance-constrained coupling mechanisms must be rejected from consideration.

Joint distribution (rejected). A more general strategy is to model the full joint distribution of state and value explicitly, for example by means of a *copula* that captures how the two vary together. A copula is a function that constructs the joint distribution of several random variables by combining their individual distributions with a separate specification of the dependence structure between them. According to Sklar's Theorem, any joint distribution can be decomposed this way and any set of individual distributions can be coupled into a joint distribution via a copula [23]. The purpose of this method is to model dependencies between uncertain quantities whose relationship is unknown and must be uncovered through statistical analysis. This is not the situation here: desirability is not a second unknown quantity that must be statistically correlated with the configuration, but rather a deterministic, specified function of it with a fixed value for the intervention being evaluated. In other words, desirability is *known*, so there is no dependence to learn. A copula formulation would therefore be unnecessary as it would expend computational resources to recover a relationship that is already known. Joint distribution methods are therefore rejected from consideration.

Bellman value-function (rejected). The most established coupling method in control theory and dynamic programming is the *value-function*, which assigns the best possible future return to each state in a system. This is defined recursively via the *Bellman equation*, which expresses a state's value in terms of the immediate reward plus the expected value of the subsequent states to which it can transition assuming the optimal path. It combines value and system dynamics by back-propagating reward through the transition structure, giving a single value for every state. This returns an optimal policy from every possible state [24]. It handles sequential decision and uncertainty rigorously for systems that should be driven automatically towards a defined target configuration. It is, however, the archetype of the perspective fusion that the proposed architecture is designed to avoid. Reachability and desirability are entangled into one singular value function from the outset, and cannot be retroactively separated for independent inspection. Additionally, it outputs an optimal policy rather than a set of options that can be

evaluated and presented to a human decision-maker for review. Value-function methods are therefore rejected from consideration.

Pushforward mapping (adopted). Given a random quantity and a deterministic scoring rule, the *pushforward* is the distribution obtained by simply mapping each value to its score, so that each resulting score inherits the probability of the value that produced it [25]. This generates a distribution applied via a function in which the random quantity is the independent variable and the score is the dependent, while preserving each's individual components without altering or fusing any information. Applied to the architecture presented in this work, each particle's state gets mapped to a desirability vector, and the calculated probability is carried to that vector, returning a probability-weighted cloud of desirability vectors. In other words, the reachability distribution is expressed through the desirability field, describing how good that intervention's attainable outcomes are over the specified horizon.

Expressed in mathematical terms, for each particle i, w_i is the weight (probability mass), and x_i is the state (point in state space), the full desirability vector D can be expressed in terms of each marker's component desirability d:

$$\mathbf{D(x_i) = (d_marker1(x_i), d_marker2(x_i), \ldots)}$$

With the pushforward mapping expressed as:

$$\mathbf{\{(D(x_i), } w_i \mathbf{)\} \text{ for all } } i$$

The resulting transformation is that the state, previously mapped to raw marker values, is now mapped to desirability scores. What is preserved is the probability weight (carried along unaltered) and separation of objectives (remain distinct components within each vector). The distributional structure and the vector structure are thus both preserved intact, no optimization is performed, and no preference is introduced. The outcome is the unbiased characterization required of the coupling mechanism. The simplicity of the mechanism is intentional, because the architecture has kept feasibility and desirability as two well-formed, independently specified objects up to this stage, combining them requires nothing more than expressing one field over samples of the other. This is the coupling mechanism selected for inclusion in the final architecture.

3.2.4 Multi-objective ranking under uncertainty

The coupling produces, for each candidate intervention, a preference-free distribution of desirability vectors. This is an objective characterization of how good that intervention's attainable outcomes are, identical regardless of who is running the algorithm or what the specific mission parameters are. *Ranking* is the operation that transforms this collection of characterizations into a comparable list that can be presented to human decision-makers to determine the proper course of action. It functions by comparing candidates'distributions, selecting the non-dominated options, and returning the short list of potential interventions to the decision-maker. This is the only stage in which subjective judgement enters the architecture, in the form of adjustable "knobs" for risk appetite and other mission preferences, defining the ranking process by specifying how heavily

unfavorable per-marker outcomes should count, and how the competing objectives should be weighed against one another.

The two modifiable inputs for weighting the ranking mechanism are **risk tolerance** and **objective priority**. Risk tolerance is the degree to which the possibility of poor outcomes should influence a candidate's assessment. The most extreme example of this case is an intervention with a 50% probability of high desirability but a 50% probability of catastrophic failure having the same average score as an intervention with a 100% probability of 50% desirable outcome, which are clearly not the same result for a mission-critical decision-maker. Therefore the "risk knob" modulates the degree to which interventions with undesirable outcomes are taken into consideration during ranking. Objective priority is the relative importance assigned to each competing objective. To use a hypothetical HSC example, mission planners may decide that for the current mission phase bone health is a higher priority than cardiovascular health, subjectively assigning higher weight to outcomes that have more highly desirable outcomes for bone mineral density preservation. Both of these inputs are supplied by the decision-maker and applied at ranking as reversible adjustments over the coupling's output, so that shifting priorities across mission phases may reshape ranking without altering the underlying distribution primitives.

The risk and priority inputs could be applied most simply by reducing each objective's marginal distribution to a scalar risk functional such as a Conditional Value-at-Risk (CVaR) or quantile threshold, then applying classical Pareto dominance to the resulting vectors. This type of mechanism is not adopted in this work in order to avoid discarding information through collapse of underlying structure. Per-objective reduction loses two things in particular: the dependence structure between objectives within a candidate intervention (whether inter-objective outcomes co-occur), and the shape of the risk distributions. However, ranking using the full distributions incurs some accepted costs, which are increased computational demand and potentially harder-to-explain rankings than a simple collapsed model. The following candidate methods are taken into consideration:

Probabilistic dominance (candidate). This runs pairwise competitions between two competitors, asking how often one beats the other. Because each protocol's outcome is uncertain, comparing two resembles simulating both and recording the more desirable protocol for every objective. The fraction of simulations in which one beats the other across the board becomes a measure of how reliably it is the better choice. The result is graded: "A beats B on objective X about 75% of the time", rather than a simple statement claiming one is better. The mathematical ranking algorithm then takes those graded results and determines the non-dominated protocols [26]. For example, a protocol can be removed from the set if it gets beaten above a threshold frequency value. The advantages of this method are that it does not require additional structure transformation of the simulation output to return a result, and it is indifferent to how each objective's desirability scale is calibrated because it only records the winner, not by how much it won. Its disadvantages are that it treats large and small differences in desirability in the same

manner (possibly underreporting disastrous results), and it does not guarantee a coherent ordering in the case of circular results (e.g., A > B, B > C, C > A).

Stochastic dominance (candidate). This method asks whether one protocol's range of outcomes is better at every outcome threshold. For example, is protocol A better than B at keeping marker X above a 1% threshold? Then ask the same question for every other possible threshold (2%, 3%, etc.). If A is at least as good as B at every single level, then we can say "A stochastically dominates B". This described condition is first-order stochastic dominance, whose verdicts hold for any type of decision-maker. The second-order condition is more relaxed, letting some negative outcomes fall behind if they are compensated for elsewhere [27]. The main advantage of this method over probabilistic dominance is that it can somewhat account for the relative differences between outcomes in the second-order condition, instead of just returning a winner. The main disadvantages are that it may frequently be unable to determine a winner in cases without a clear better option, by considering irrelevant risk tolerances for the particular case. Another disadvantage is that when applied per-objective, it ignores inter-objective outcome co-occurrence, which was one of the motivating factors for keeping the full distribution intact in the first place. Also, it negates the need for a modifiable risk tolerance input, as it covers all risk-averse utilities in its ranking algorithm (which is neither a categorical advantage nor disadvantage).

Neither of the comparison methods dominates on all relevant properties, so the decision on which shall be implemented in the final architecture is deferred to a later comparative evaluation. The experimental evaluation will use synthetic distributions with differing structures to assess relative performance on the following list of criteria:

- computational cost
- stability of the returned set under Monte Carlo resampling and varying sample size
- the size of the returned set and the incomparability rate
- response to changes in the two modifiable inputs (risk tolerance and objective priority)
- agreement among the candidates and compared to other benchmark methods
- interpretability of results

Aggregating comparisons (adopted). Whichever comparison method is adopted, its output is a relation between pairs of protocols, and not the set of non-dominated protocols the architecture is supposed to return. A further step is therefore needed to turn pairwise comparisons into a presented set. The output of comparison is coarse and incoherent to a human operator. Aggregation resolves this by converting pairwise verdicts into a per-protocol ranking. The rule adopted here is that for each protocol, sum over all competitors the probability that the competitor dominates it, and present the protocol only if that total falls below a certain threshold [28]. A protocol with a low total is one that is unlikely to be beaten, a high total is probably a poor choice. This construction can be input to a ranking index called the *Pareto Uncertainty Index* (PUI) [29], with three properties that recommend it for adoption in the architecture: the threshold can be directly proportional to the risk appetite, it returns a graded set instead of a binary one, and it can

simultaneously return a wider set of solutions near the non-dominating frontier and a concrete "short list" of the best solutions directly on it. The limitations associated with this technique are that each protocol's score depends on which other protocols are in the pool it is competing with (scores are not comparable across different pools), with small pools the normalized score has few values (possibly leaving protocols unseparated), and the aggregation inherits any "blindness" from the comparison method (e.g., under probabilistic dominance, the severity of bad outcomes is still unseen).

3.2.5 Model personalization

Both the feasibility and desirability models depend on individualization of the representation if the architecture is to power a digital twin. The feasibility model asks what states a given protocol can drive *this system* toward, and the desirability model asks how good each state is *relative to the individual's own reference values*. These can both be informed by baseline measurements before any modeling begins, but must be kept updated as dynamic systems change their configuration over time (as result of an implemented protocol or spontaneously). Two properties are required of the personalization paradigm: it must carry uncertainty rather than collapse it, and it must update as new evidence arrives. The three paradigms presented below are assessed against these requirements.

**Note that the shared paradigm between the two layers does not imply shared inputs, as it is imperative that they are specified independently.*

Point-estimate fitting (rejected). The simplest personalization fits the individual's parameters once (e.g., from preflight baseline measurements for astronauts) and uses the resulting best-fit values thereafter, with periodic refitting as conditions permit. Estimation is by standard curve fitting or maximum likelihood, and the result is a single parameter vector describing the system. The advantages are the cheap computational cost, the small amount of data needed to make an estimate, and the direct interpretability/auditability of results. However, its simplicity makes it a less faithful model of the physical twin, and directly fails to differentiate individuals besides in the most trivial manner. It can be rejected outright because it is not robust enough to model complex systems.

Filtered sequential estimation (rejected). This method maintains a running estimate of the individual's state and parameters and revises it whenever a new measurement arrives, carrying a covariance alongside the estimate. The Kalman filter is the standard technique here, with its extended, unscented, and ensemble variants accommodating progressively more nonlinear system dynamics [18]. It appears to satisfy the two requirements cleanly: it carries uncertainty, and the estimate tightens as evidence accumulates. Additionally, the computational cost is relatively low, making this a reasonable option for unimodal systems with roughly symmetric structures and dynamics. Its disadvantages, however, render it a poor fit. The uncertainty it assumes is Gaussian in shape and does not represent multimodal data structures well. The architecture is specifically slated to work on complex, multimodal, irregular data shapes so it needs a personalization

paradigm that fits those goals. A marker in which the individual may occupy one of several distinct response regimes may be represented by this method as in between regions, unfaithful to the data. This paradigm is therefore rejected in favor of a more robust option.

Hierarchical Bayesian individualization (adopted). This paradigm maintains a full posterior distribution over the individual's parameters, updated sequentially as new evidence arrives. The individual is also modeled as a member of a cohort, so that in the absence of individual data the estimate defaults to known variation, and moves towards personalized dynamics as measurements accumulate. Two features distinguish it from filtered sequential estimation: the posterior may take any shape, and the cohort level supplies an explicit prior. Both stated requirements are met, with uncertainty faithfully matching the individual's real dynamics (especially after many updates), and the estimates tightening as new data accumulate. Bayesian individualization comes with some real costs, which should be mentioned. First, it can be substantially more computationally costly than the other paradigms, especially if sequential Monte Carlo simulation is used. Second, the prior is be dependent on cohort data, which may be small (especially in the case of astronauts, which have notoriously small cohorts). Third, a Bayesian posterior is less interpretable than a simple number (like in the case of point-estimate fitting). Still, because of its robustness and usability with complex data structures, Hierarchical Bayesian individualization is the paradigm chosen to be used in the architecture.

3.3 Closest prior work

This section presents a consolidated list of the closest published systems, found in the literature, to the architecture presented in this work. These are end-to-end algorithms that occupy a similar niche, from which this work must prove its distinctness. The systems presented here are valuable tools for determining the state-of-the-art within the personalized decision-support algorithm niche, analyzing the respective differences of each, and positioning this work's architecture within the field.

Predictive digital twin for optimizing patient-specific radiotherapy regimens under uncertainty in high-grade gliomas (Chaudhuri et al., 2023) [30]***:*** This published methodology is the closest to this architecture, with many components similar in scope to the ones adopted in this architecture. Their algorithm seeks to accomplish a very similar goal, but with some important differences in terms of mechanism. Both architectures address critical decisions in which individual response varies significantly, and act as decision-support algorithms returning a set of presented options for a human to choose from with different levels of trade-off, in multi-objective cases with genuinely conflicting objectives. There are several shared mechanisms, with the present work adopting some of the same components. Their digital twin is initialized from population-level clinical data, then personalized via Bayesian calibration assimilating patient-specific data. This is the same paradigm chosen for this work's personalization paradigm. In the Chaudhuri et

al. architecture, forward propagation of uncertainty is also done via Monte Carlo simulation to realize the outcome distribution.

The two architectures, however, diverge in several loci. In Chaudhuri et al., the outcome distribution for tumor control is reduced to a single risk-adjusted scalar inside the objective function, and that scalar is minimized against a constraint on total radiotherapy dose to trace a trade-off curve. The collapse of component information therefore occurs before any comparison between candidate interventions is made, unlike in the present architecture where the coupling stage yields a full, unbiased probability distribution of desirability vectors where it is carried intact to the ranking stage. Three distinct conceptual differences follow: dependence between objectives is preserved rather than discarded by per-objective reduction, risk posture is a property of the final ranking instead of as part of search allowing for computationally cheap reranking with different preferences, and the desirability model is specified independently rather than being a function of the trajectory inside an optimization problem.

Dose-finding based on efficacy–toxicity trade-offs (Thall & Cook, 2004) [31]***:*** The idea of scoring outcomes against a value field that is specified separately from the model that predicts them is not completely novel in clinical decision-making. Thall and Cook (2004) constructed, for phase/II dose-finding, a desirability field defined over the entire two-dimensional domain of possible efficacy and toxicity probabilities. The field is built by eliciting three outcome pairs judged equally desirable, fitting a contour through them, and generating a family of contours such that all outcomes on one are equally desirable. Nothing in its construction refers to the dose-response model; it scores any outcome that the model might produce. Their method also evaluates a discrete, physician-specified candidate set, admits a dose only if it exceeds certain specified thresholds, and grounds its priors in elicited expectations. There are some important differences. First, their desirability field returns a scalar value, so trade-offs are resolved in the field itself. Second, their ranking algorithm evaluates candidates by desirability of the posterior mean, thereby collapsing the distribution and returning a single selected dose.

A survey of multi-objective sequential decision-making (Roijers, Vamplew, Whiteson and Dazeley, 2013 [32]***).*** The decision-theory concept adopted in this work is not completely novel, Roijers, Vamplew, Whiteson and Dazeley (2013) identify three scenarios in which converting a multi-objective sequential decision problem into a single-objective one is impossible, infeasible, or undesirable, one of which called the *decision support scenario* describes the present setting. In their formulation, scalarization is infeasible throughout the entire decision-making process in order to not lose information early, since different stakeholders may have preferences that require separation of components, or their preferences may change at a later stage. In these cases, early scalarization forces users to express their preferences unnaturally, and the ideal workflow in decision analysis should present alternatives *before* soliciting user preferences. Some other similarities include that the returned set not contain “suboptimal” (referred to as “dominated” in this work) options, and that it is best to not assume you know the decision-maker’s preferences when designing the model and to leave it more user-customizable in terms of objective priorities.

4. Conclusion

This report has specified an algorithmic architecture for decision support in intervention selection: the problem of choosing, for an individualized system under uncertainty, which of a set of candidate interventions to apply when the objectives at stake genuinely compete and the results of the decision are mission-critical. Its organizing commitment is the separation in scope of reachability and desirability, independently informed from separate evidence, and combined only when a decision-maker needs to take action.

Five design decisions mold the computational form. Desirability is a graded, vector-valued field defined over the whole state space, never aggregated into an index. Feasibility is a forward, purely descriptive operator returning a weighted particle representation of reachable states, adopting no target and computing no optimum. Coupling is a pushforward, the minimal operation that expresses one layer over samples of the other while preserving both the distributional and vectorial structures. Ranking is the single stage at which subjective judgement enters, through two reversible user inputs applied over stored results rather than inside a search. Personalization uses a Hierarchical Bayesian methodology to faithfully represent complex uncertainty shapes. The final output is a non-dominated set presented to a human decision-maker.

The architecture's novelty is one of composition rather than de novo invention. Its components are drawn from control theory, stochastic multi-objective optimization, Bayesian inference, and digital twin modeling. What the literature does not yet contain the conjunction of independent specification, query-time coupling, and evaluation rather than optimization. Additionally, the characterization of the problem class for which this architecture is designed is not specifically referenced in the literature.

This report establishes that the architecture is specified at a level sufficient for implementation, that each of its components is justified against the alternatives, and that the promised decision-support output follows from its construction. The next phase is creating a proof-of-concept software implementation of the architecture, and experimentally validating it against clinical data.

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